

3D Printing and Modeling

Final Project Requirements

1. Description

Dear students, the time has come for you to deliver a final project. You must create your own, finished product, keeping in mind that it needs to be printable. The product can contain non-3D printed parts to your discretion, but must remain primarily 3D printable.

2. Requirements

The total grade of 10 is split into the following:

- Design Quality - **5 Points**
- Movement - **1 Point**
- Printability - **2 Points**
- Publishing - **2 Points**

3. Design Quality

Design Quality refers to the quality of your work from start to finish. Your task is to create ***your own*** product starting from any inspiration sources you wish. You are not allowed to simply copy an existing design. Your design must serve a purpose, make sense and you must be able to explain how it works and why it works. When starting from a design already found online, you must be able to articulate what your contribution is and how you approached redesigning it.

Design Quality also refers to your work inside of **Fusion 360**. You must work as cleanly as possible, following the same set of rules you did for homework. (Naming your sketches, organising your components, etc.)

4. Movement

As mentioned before, your project ***MUST*** include movement in its design. It needs to have at least one motion link and one motion study to animate your product.

5. Printability

As the name suggests, it refers to how easy it is to **3D print your product**. You do not have to print your project, but you must consider how it could be printed when designing it. Make sure you break up your model into components / bodies that can each be 3D printed. Try designing your components so they require **as little support as possible** when printed. Chamfer / fillet your edges, consider printing orientation for strength, include appropriate tolerances, split larger bodies that might not fit on a build plate etc.

Obviously, not every component can be printed without supports, but you will lose points for components that could have easily been designed better to fit these requirements.

6. Publishing task: Github ReadMe

For the final project, you must create a new, public github repository. This repository will contain all the details of the project, similar to a journal detailing how you approached the task.

It **MUST** contain:

1. Detailed project description;
2. All project files used; (.f3d project, STL file(s), g code files for all printable components)
3. Link to a video showcasing functionality (record your screen while playing your motion links, showing how it could be assembled, etc. Upload ideally to **Youtube**);
4. A list of components / parts used, including their purpose;
5. List of resources used;
6. Anything else you believe to be necessary;

As there is no list of projects from last year, I will leave a few example repos from this year's "*Introduction to Robotics*" course. Evidently, not everything in these pages will apply to your project (ex: you do not need to write any code), so treat them more as inspiration than templates:

- [Introduction to Robotics](#)
- [Robotics Crane Project](#)
- [Color-Sorting Trash Bin](#)
- [We got a Space Problem](#)
- [Productivity Bloom](#)

If you are unfamiliar with creating your own Readme, the best approach is to start by copying someone else's. Here's a [list](#) of existing github projects, each

with a decent Readme file. You can [clone](#) any of these and open Readme.md in your own text editor to see how it was created.

7. Deadlines

20.05.2026 - Link to your repository in the assignment.

02.06.2026 - Final commit.